Relationship

$P_{25} = Q_1$

P_{50} = Median = Q_2

P_{75} = 3rd quartile = Q_3

Example 5.35

The following is the monthly income (in 1000) of 8 persons working in a factory. Find P_{30} income value

10,14, 36, 25, 15, 21, 29, 17

Solutions:

Arrange the data in an ascending order.

$n = 8$

10,14,15,17,21,25,29,36

$$P_{30} = \left(\frac{30(n+1)}{100}\right)^{th} \text{ item}$$

$$= \left(\frac{30 \times 9}{100}\right)^{th} \text{ item}$$

$= 2.7^{th}$ item

$= 2^{nd}$ item + 0.7 (3rd items – 2nd items)

= 14+0.7(15-14)

= 14+0.7

P_{30} = 14.7

Example 5.36

Calculate P_{61} for the following data relating to the height of the plants in a garden

Heights (in cm)	0 – 5	5 – 10	10 – 15	15 – 20	20 – 25	25 – 30
No of plants	18	20	36	40	26	16

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